

Minimum Wages and Employment: Replication of Card & Krueger (1994)

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1. Research question and motivation

Does raising the minimum wage reduce employment? Standard competitive-labour-market theory predicts that a binding wage floor makes hiring more expensive and thus lowers employment. Card & Krueger (1994) test this prediction using a natural experiment: on **April 1, 1992, New Jersey raised its minimum wage from \$4.25 to \$5.05**, while neighbouring **Pennsylvania kept it at \$4.25**. Because the two states are adjacent and economically similar, the New Jersey change provides a plausibly exogenous shock to the cost of low-wage labour, and Pennsylvania serves as the control group.

The paper is a landmark application of the **difference-in-differences (DiD)** design and remains a canonical reference for policy evaluation. Its headline result — that fast-food employment in New Jersey did **not** fall (and, if anything, rose) relative to Pennsylvania — challenged the textbook view and motivated a large follow-up literature. This project reconstructs that result from the original survey data.

2. Data, sample construction, and variable definitions

Source. Two-wave telephone survey of fast-food restaurants in New Jersey and eastern Pennsylvania (Burger King, KFC, Roy Rogers, Wendy's). Wave 1 was conducted February–March 1992 (pre-treatment); Wave 2 was conducted November–December 1992 (post-treatment). The raw file data/raw/public.csv contains **410 restaurants × 46 columns**, one row per restaurant, with Wave-1 variables (no suffix) and Wave-2 variables (suffix 2) side by side.

Sample construction. - **Store identifier.** The questionnaire number SHEET is not unique (number 407 labels two different restaurants), so the row index is used as the store id. - **Balanced sample.** The DiD uses restaurants observed in both waves: 384 stores for employment (309 NJ, 75 PA).

Key variables.

Variable	Definition
STATE	1 = New Jersey (treatment), 0 = Pennsylvania (control)
treated	= STATE
EMPFT / EMPPT / NMGRS	full-time, part-time, and manager counts

Variable	Definition
FTE	full-time-equivalent employment = EMPFT + 0.5 · EMPPT + NMGRS
WAGE_ST	starting wage
CO_OWNED	1 = company-owned, 0 = franchise
CHAIN	brand (1=Burger King, 2=KFC, 3=Roy Rogers, 4=Wendy's)

The FTE definition and the STATE coding were validated by matching the computed Wave-1 means to the paper's published values (see §4) — an exact match.

3. Empirical design

The causal effect of the minimum wage is estimated by double differencing:

$$DiD = [E(NJ, \text{Wave } 2) - E(NJ, \text{Wave } 1)] - [E(PA, \text{Wave } 2) - E(PA, \text{Wave } 1)]$$

Two equivalent implementations are used: 1. a comparison of mean changes across the two states (the paper's Table 3), and 2. reduced-form regressions of the store-level change on treatment and controls (the paper's Table 4),

$$\Delta Y_i = \alpha + \delta \cdot NJ_i + \varepsilon_i,$$

where δ is the DiD estimate. Table 4 reports ordinary (homoskedastic) OLS standard errors, as in the paper; the ownership extension and the robustness checks use HC1-robust standard errors. The first stage — that the policy actually raised New Jersey wages — is confirmed directly by the starting-wage means in Table 2 (New Jersey: \$4.61 → \$5.08; Pennsylvania: flat at \$4.62), and the employment version is the *main result*. The identifying assumption is parallel trends: absent the policy, NJ and PA employment would have moved together.

4. Core replication results and comparison with the original

Table 2 — Means of key variables. Table 2 reports descriptive statistics: the distribution of store types (Panel A) and the means of the key variables in Wave 1 (Panel B) and Wave 2 (Panel C), separately for New Jersey and Pennsylvania. Following the paper, the employment-composition variable is the *percentage of full-time employees* (the store-level mean of $100 \times \text{EMPFT}/\text{FTE}$), not the raw staff counts.

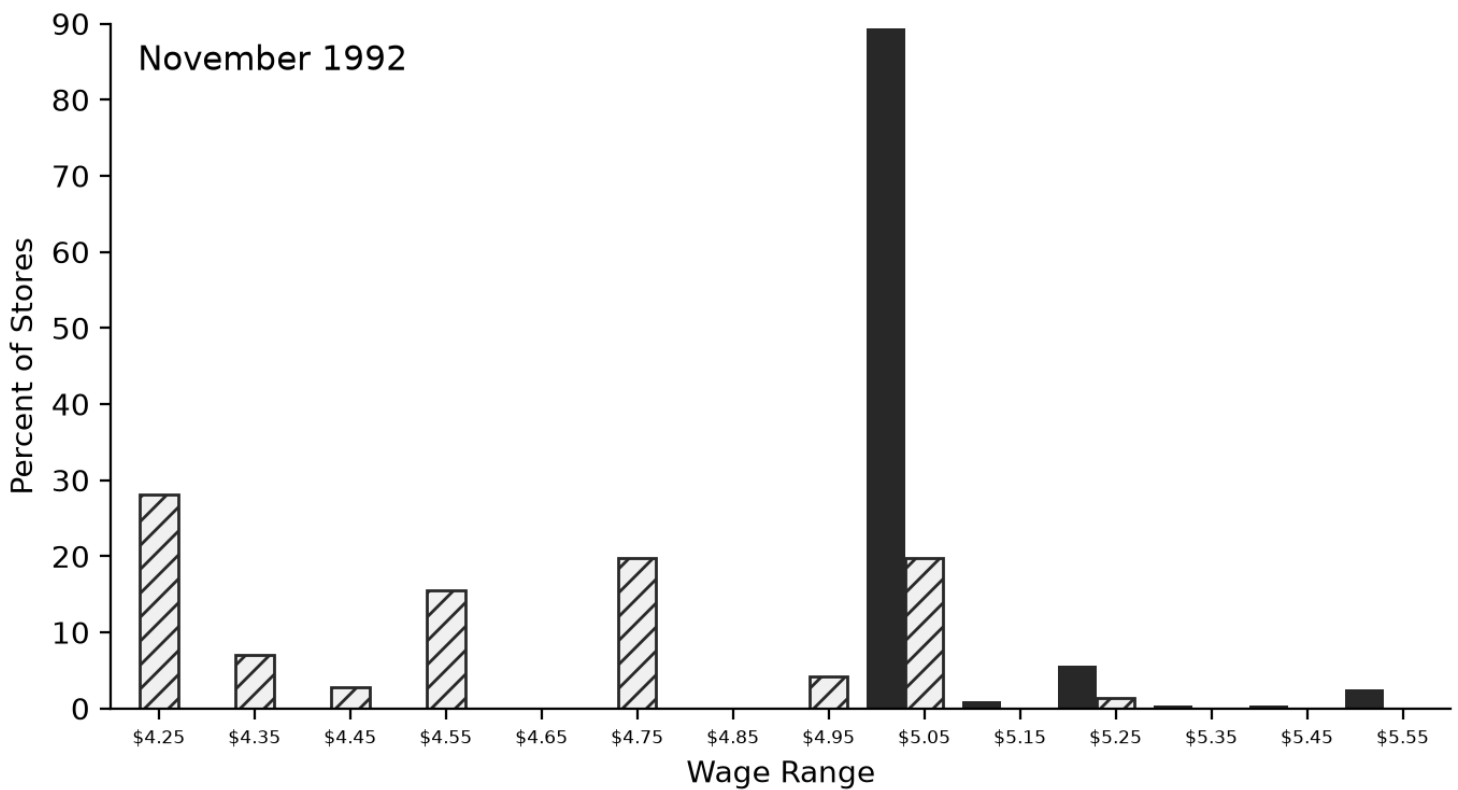
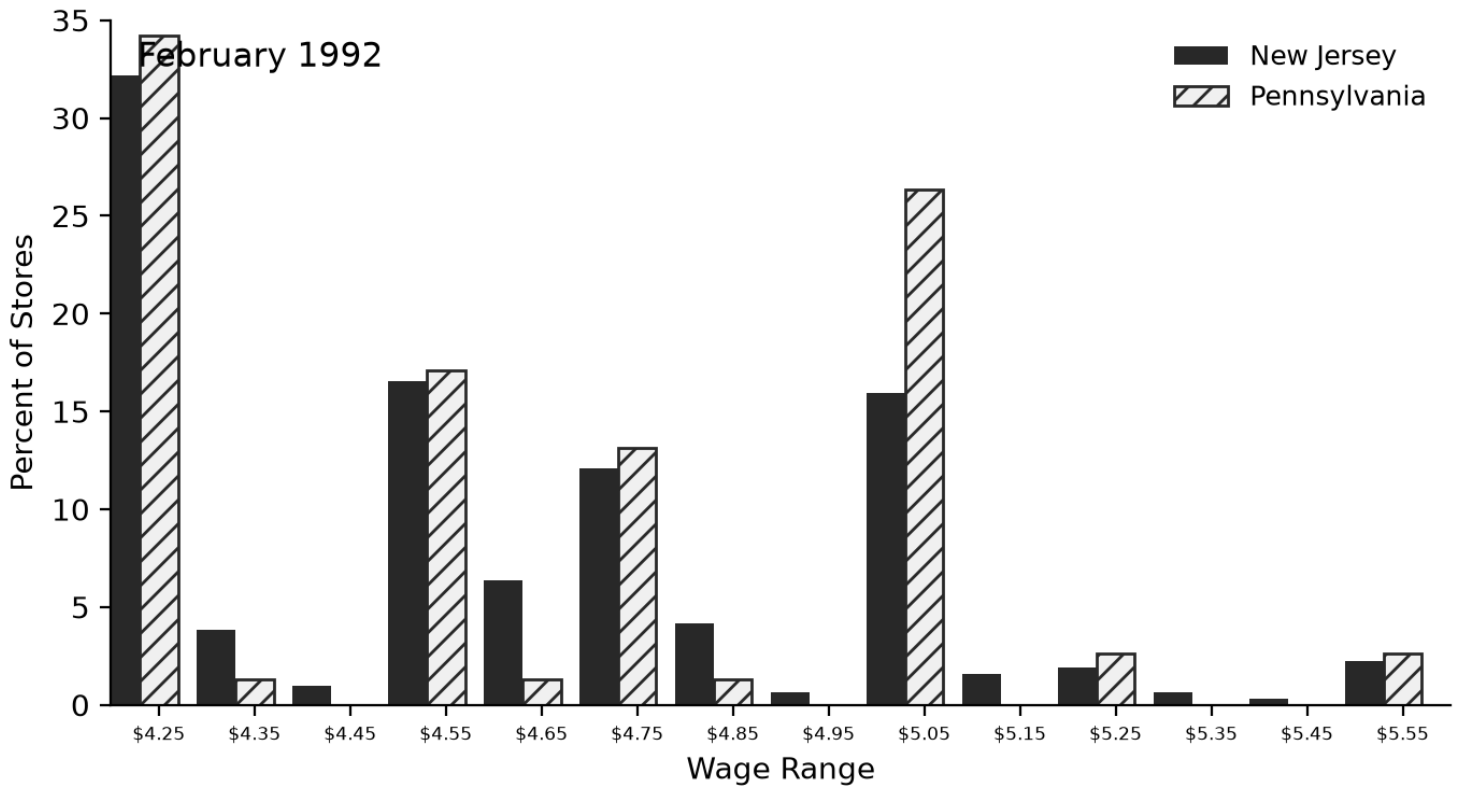
Panel	Variable	NJ	PA
A. Store types (%)	Burger King	41.1	44.3

Panel	Variable	NJ	PA
	KFC	20.5	15.2
	Roy Rogers	24.8	21.5
	Wendy's	13.6	19.0
	Company-owned	34.1	35.4
B. Wave 1 means	FTE employment	20.44	23.33
	Percentage full-time employees	32.8	35.0
	Starting wage (\$/hr)	4.61	4.63
	Wage = \$4.25 (%)	30.5	32.9
	Price of full meal (\$)	3.35	3.04
	Hours open (weekday)	14.42	14.53
	Recruiting bonus (%)	23.6	29.1
C. Wave 2 means	FTE employment	21.03	21.17
	Percentage full-time employees	35.9	30.4
	Starting wage (\$/hr)	5.08	4.62
	Wage = \$4.25 (%)	0.0	25.3
	Wage = \$5.05 (%)	85.5	1.3
	Price of full meal (\$)	3.41	3.03
	Hours open (weekday)	14.42	14.65
	Recruiting bonus (%)	20.3	23.4

The store-type mix is essentially identical across the two states, confirming their comparability. The two waves show the policy's bite: New Jersey's average starting wage jumps from \$4.61 to \$5.08 while Pennsylvania's stays at \$4.62, and the share of New Jersey stores paying the new \$5.05 minimum reaches 85.5%.

Figure 1 — Distribution of the wage rate. Figure 1 plots the distribution of starting wages in two stacked panels — February 1992 (top) and November 1992 (bottom) — each grouping New Jersey and Pennsylvania bars side by side, with the y-axis in **percent of stores**. Following the paper, the wage bins run in \$0.10 increments from \$4.25 to \$5.55 (fourteen bins); each panel has its own y-axis scale (February 0–35, November 0–90) and its own "Wage Range" x-axis, with the tick values labelled at every \$0.10.

Figure 1 — Distribution of Starting Wage Rates



Before the reform both states' wages cluster at the old \$4.25 minimum; after the reform the New Jersey distribution shifts right to the new \$5.05 minimum while Pennsylvania stays at \$4.25. The February New

Jersey \$4.25 bar holds 32.2% of stores and the November New Jersey \$5.05 bar holds 89.3%, matching the paper's figure. Stores reporting a wage above \$5.55 (including the single Pennsylvania store at \$6.25 in November) fall off the right edge of the axis and are not plotted.

Table 3 — Average employment per store before and after. Table 3 is the paper's headline table, and it reports **FTE employment only**. Columns (i)–(ii) give Pennsylvania and New Jersey, column (iii) the NJ – PA difference; columns (iv)–(vi) split New Jersey by the wave-1 starting wage (\$4.25, \$4.26–4.99, $\geq$ \$5.00), and columns (vii)–(viii) give the low–high and midrange–high contrasts.

FTE employment	PA	NJ	NJ – PA	\$4.25	\$4.26–4.99	$\geq$ \$5.00	Low – High	Mid – High
Wave 1 (before)	23.33	20.44	–2.89	19.56	20.08	22.25	–2.69	–2.17
Wave 2 (after)	21.17	21.03	–0.14	20.88	20.96	20.21	0.66	0.74
Change (unbalanced)	–2.17	0.59	2.75	1.32	0.87	–2.04	3.36	2.91
Change (balanced sample)	–2.28	0.47	2.75 (SE 1.34)	1.20	0.71	–2.16	3.36	2.87
Change (temporarily closed $\rightarrow$ 0)	–2.28	0.14	2.42	0.61	0.49	–2.39	3.00	2.88

The balanced-sample difference-in-differences is **+2.75 FTE employees** (SE 1.34, $t = 2.05$), reproducing the paper's +2.76 ($t = 2.03$). Within New Jersey the same pattern holds: low-wage stores grew (+1.20) while high-wage stores shrank (–2.16), for a low–high gap of +3.36 and a mid–high gap of +2.87 — the monotone gradient the paper documents.

Table 4 — Reduced-form models for change in employment. Table 4 estimates the employment effect in regressions of the store-level change in FTE employment on a New Jersey dummy (models (i)–(ii)) or on GAP, the proportional wage increase a store needed to reach the \$5.05 minimum (models (iii)–(v)), with controls added progressively. Standard errors (in parentheses) are the ordinary (homoskedastic) OLS errors, as in the paper.

Independent variable	(i)	(ii)	(iii)	(iv)	(v)
New Jersey dummy	+2.36 (1.15)	+2.33 (1.15)	—	—	—
Initial wage gap	—	—	+15.89 (5.95)	+15.19 (6.08)	+12.03 (7.32)
Controls for chain and ownership	no	yes	no	yes	yes
Controls for region	no	no	no	no	yes
Standard error of regression	8.71	8.70	8.68	8.67	8.67
Probability value for controls	—	0.299	—	0.381	0.375

Notes. Standard errors are given in parentheses. The sample consists of 365 stores with available data on employment and starting wages in waves 1 and 2 (the paper reports 357); the dependent variable in all models is the change in FTE employment (mean −0.252 and SD 8.749 here, vs −0.237 and 8.825 in the paper). All models include an unrestricted constant (not reported). $GAP = (5.05 - WAGE_ST) / WAGE_ST$ for New Jersey stores initially paying below \$5.05, and zero otherwise. The chain-and-ownership controls are three chain-type dummies plus a company-owned dummy; the region controls are dummies for the three New Jersey regions (South, Central, North) and two eastern Pennsylvania regions, one omitted as the reference; and the last row reports the p-value of the joint F test for exclusion of all control variables.

The positive GAP coefficient has the expected gradient interpretation: stores facing the largest mandated wage increase saw the largest employment gains, and the estimate is robust to controls. Adding region controls (column (v)) attenuates the coefficient and raises its standard error, mirroring the paper. The estimates track the paper's (2.33, 2.30, 15.65, 14.92, 11.91); the small residual differences reflect the public dataset's sample (n = 365 vs the paper's 357).

Comparison with the original paper. The paper reports an FTE-employment DiD of **+2.76 (SE 1.36)**. Our estimate — **+2.75 (SE 1.34)** — reproduces it essentially exactly, and the wave-1/wave-2 means match the published values to two decimals. The reduced-form GAP estimates (+15.89 vs the paper's +15.65) and the New Jersey dummy (+2.36 vs +2.33) also track the original. The substantive conclusion is unchanged: New Jersey's minimum-wage increase did **not** reduce employment. Relative to Pennsylvania, FTE employment rose by about 2.75 workers per store, with the share of full-time staff rising in New Jersey relative to Pennsylvania (32.8 → 35.9% vs 35.0 → 30.4%).

5. Independent extension: does the effect differ by store ownership?

Question. Is the (null/positive) employment response uniform, or does it differ between **franchise** and **company-owned** outlets, which face different cost structures, compliance, and local market power?

Design. On the balanced sample, estimate

$$\Delta FTE_i = \alpha + \delta \cdot NJ_i + \beta \cdot CO_OWNED_i + \gamma \cdot (NJ_i \times CO_OWNED_i) + \varepsilon_i,$$

so δ is the DiD for franchises, $\delta + \gamma$ for company-owned stores, and γ the difference.

Results.

Group	NJ change	PA change	DiD
Franchise (CO_OWNED=0)	+0.72	−2.41	+3.13 (SE 1.89)
Company-owned (CO_OWNED=1)	−0.01	−2.05	+2.04 (SE 1.53)

Group	NJ change	PA change	DiD
Difference (interaction)			-1.10 (SE 2.43, p = 0.65)

Interpretation. The positive employment effect appears in *both* ownership types, and the interaction is far from statistically significant. There is therefore no evidence that the minimum-wage effect is driven by, or concentrated in, a single ownership type — the headline result is robust across ownership structures. The limited precision (small Pennsylvania cells of 49 and 26) is noted as a caveat.

6. Validation, limitations, and conclusion

Validation. - The FTE formula and state coding reproduce the paper's Wave-1 means exactly (NJ 20.44 / PA 23.33 for employment; NJ 4.61 / PA 4.63 for wages). - DATE2 (MMDDYY) spans 11-05-1992 to 12-31-1992, matching the documented Wave-2 field period. - The duplicate SHEET number and the NJ-region-dummy collinearity were identified and handled (see the AI-use disclosure).

Robustness. The employment DiD is stable across specifications. On the full balanced sample the baseline is +2.75 (Table 3); restricting to stores with complete controls (n = 361), it is +3.02 with no controls, +2.98 adding chain fixed effects, and +2.70 adding chain FE, ownership, hours, and prices. All are positive, and the first three are statistically significant at conventional levels.

Limitations. (i) The Pennsylvania control group is small (75 stores in the balanced sample), limiting precision; (ii) with only two waves, the parallel-trends assumption cannot be formally tested; (iii) the outcome is a snapshot measure of employment in a single low-wage industry, so external validity is limited.

Conclusion. The replication recovers the paper's famous finding almost exactly: New Jersey's 1992 minimum-wage increase did not reduce fast-food employment and, relative to Pennsylvania, employment rose by roughly 2.75 FTE workers per store. An original extension finds no significant heterogeneity by ownership type, reinforcing the robustness of the result.

Reproducibility

With Python 3.14 and the packages in `requirements.txt` (pandas, numpy, statsmodels, matplotlib, markdown), run from the project root:

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -I code/run_all.py
```

This regenerates all tables in `outputs/tables/` and figures in `outputs/figures/`.